

open syllable: syllable ends w/ a vowel

Phonological Env

∅ null

\$ syllable boundary

word boundary

pronounce $\begin{bmatrix} \text{hue} \\ \text{guava} \end{bmatrix}$ as $\begin{bmatrix} \text{'ya'} \\ \text{'wa'} \end{bmatrix}$, transcribe as $\begin{bmatrix} /j\text{u}/ \\ /w\text{a}/ \end{bmatrix}$ not $\begin{bmatrix} /i\text{u}/ \\ /u\text{a}/ \end{bmatrix}$

Phone: [b], [p], [m], etc

Phoneme: 'maf', 'paf' → written as /b/, /m/
minimal pairs

Allophone: A phone that is a phoneme
e.g. [t], [t^h] are allophones of /t/

Distributions of phones

- Complementary
 - ↳ never in same phonological env
 - ↳ [t], [t^h]
 - ↳ cannot form minimal pairs
- Contrastive
 - ↳ same env, change meaning
 - ↳ "tent", "dent"
- Free variation
 - ↳ same env, does not change meaning
 - ↳ [then] vs [ɜ̃hen]

Phonological Rule

/t/ → [ɔ] / [t bilabial] —
least restrictive phoneme allophone position determiner env sound of interest

tap	[tap]	'yes'
muN	[mun]	'dog'
muNga	[munga]	'dogs'
zumat	[zumat]	'tree'
zunatka	[zunatka]	'trees'
punbesa	[punbesa]	'water'
seNbunbesa	[sejbunbesa]	'waterfall'
voniz	[voniz]	'book'
vonizka	[vonizka]	'books'
kumfam	[kumfam]	'child'

mensip	[mensip]	'bread'
papan	[papan]	'happy'
peltan	[peltan]	'car'
peltanga	[peltanga]	'cars'
tambeltan	[tambeltan]	'truck'
ketja	[ketja]	'horse'
tuNbind	[tunbind]	'factory'
zemdov	[zemdov]	'friend'
temguk	[temguk]	'love'
indemguk	[indemguk]	'lovers'

/t/ tap front open
 zumat front open
 zumatka voiceless velar plosive
 peltan front open
 voiced alveolar liquid
 tambeltan front close-mid
 tunbind back close
 temguk close mid-front

/p/ tap back closed
 punbesa front closed
 mensip front closed
 papan front open
 peltan front closed mid

/g/ mugga voice nasal velar front open
 pertangan front open
 fengale front open
 voiced bilabial nasal
 /b/ punbesa front semi-high
 sungun besa back closed
 tunbind front closed
 voiced nasal velar

/k/ zumatka front open
 voiceless plosive alveolar
 vonizka front open
 voiced alveolar fricative
 kumfam back closed
 ketja front close-mid
 fenguk back closed
 /d/ tunbind voice alveolar nasal
 zemdov back close-mid
 voiced bilabial nasal

indemguk closed mid-front
 voiced alveolar nasal

Morphology

Inflectional : Adds grammatical information but same word class

e.g. person, number, tense

Derivational : Changes word class & "adds an entry in dictionary"

boosters

breakthrough

asymptomatic

Syntax

constituent: A single unit in the sentence

As long as any **one** tests is positive, it is a **constituent**

Tests:

1. Topicalisation

↳ move to front of sentence

2. Cleft/pseudocleft ^{→ move to subj}

↳ Add words like "that"

3. Substitution

4. Deletion

Noun-Phrase Rules

Consider **phrases** not words

$$\underset{\text{noun phrase}}{NP} \rightarrow (det)(adj)^* N \underset{\text{prepositional phrase}}{(PP)}$$

Argument: arg for a verb

Adjunct: additional info

Indirect obj:

Direct obj:

Transitive: Requires an obj(?)

Intransitive: No arg/obj required

Ditransitive: Requires 2 obj (direct + indirect)

Semantics

2 aspects of meaning:

- Reference — referent in real world
- Sense — how it refers

Non-referring expression has no real-world referent

Constant reference: same referent regardless of when/where
↳ e.g. speed of light

Variable reference: depends on context
↳ e.g. the lights

The homos:

homophones — pronounced same way

homographs — spelt same way

homonyms — both homophones & homographs

Polysemes — similar to homonyms but the
2 words have some relation

hyponym (under) — NUS is a hyponym of university

hypernym (over) — School is a hypernym of university

Synonyms — approx. same meaning

Antonyms — opposing meanings

Grabable (continuum) — short-tall
Complementary (category) — alive-dead
Relational (w/ each other) — parent-child

entailment: A entails B if $A \Rightarrow B$
(implies)

contradicts: $A \oplus B \parallel A = B'$
(xor)

paraphrase: $A \Leftrightarrow B$
(iff)

presuppose: B is on implicit background assumption of A
 $(A \Rightarrow B) \wedge (A' \Rightarrow B)$
(implies regardless)

keep proposed sentences simple to avoid logical errors

Pragmatics

Expressive

Commissive - Committing to an action

Representative - Representing a belief, not necessarily true/false

Directive - Direct/instruct addressee to do something

Gricean Maxims

- Quality: X false $\Rightarrow$ sarcasm? Usually when we flout a maxim,
 $\hookrightarrow$ generates extra meaning
= conversational implicature
- Quantity: Amt of info
- Relation: Be relevant
- Manner: X ambiguity

Politeness theory

face : to want to be liked $\rightarrow$ and criticism/praise

-ve : to want autonomy

Dialect: Phonology, lexicon, syntax

Accent: Phonetics & Phonology

Dialect vs language

Mutual intelligibility $\Rightarrow$ Dialect

$\hookrightarrow$ both ways

Sociolinguistic factors

Standard vs non-standard:

- Standard varieties — intelligent, wealthy, educated (status)
- Non-standard — friendly, trustworthy (solidarity)

Social variation: 1. diff speakers, 2. speaker

Stylistic variation:

Midterm Notes

Phonology

Minimal pair:

- ↳ Change 1 **phone** to get another word
- ↳ e.g. **seed** vs **sued**

The realizations of a **phoneme** are called **allophones**

Phonological Rules:

A (Phoneme) → **B** (Allophone) / $\left[\begin{smallmatrix} + \text{plative} \\ - \text{voiced} \end{smallmatrix} \right]$ env $\rightarrow$ #

"A is realized as B" when...

↑
phoneme of concern (A)

\$ = syllable boundary
= word boundary
∅ = null

Phonological Processes:

Assimilation

- ↳ 2 sounds sound more similar
- ↳ e.g. **In** + **replaceable** = **Inreplaceable**

Dissimilation

- ↳ e.g. reducing /r/ so easier to pronounce

Epanthesis

- ↳ Phone Added

Deletion

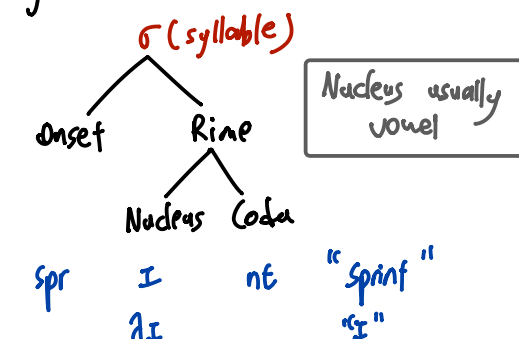
- ↳ Phoneme not realized
- ↳ e.g. **went** → [wen] in S6 English

Metathesis

- ↳ Sounds switched
- ↳ e.g. /r/ in **iron** moved to end in [aɪərən]

Synchronic: Phoneme change in same phonological env
Diachronic: Phoneme change over time

Syllable Structure



- syllable boundary
- primary stress aft syllable
- secondary stress

Morphology

- Meaning X pronunciation
- **Morpheme**: smallest unit of language

Affixes

Prefix	pre-	re-
Suffix	-ing	-ist
Infix	-s-	(e.g. passersby)
Circumfix	ke-	-an (Malay)

} uncommon in English

Productive Affix: New words easily created

Unproductive Affix: e.g. -lock

Inflectional Affix: Grammatical, const class

Derivational Affix: Δ in word class/meaning

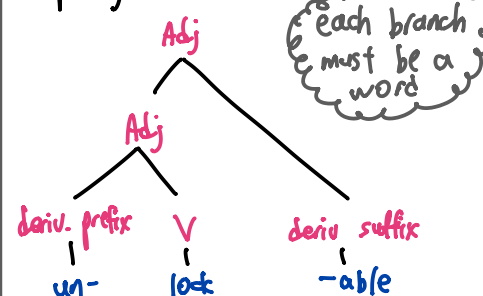
Allomorphs: Alternate forms/pronunciation of same morpheme
e.g. in-, im-

Bound morpheme: Cannot exist on its own
ind. affix

Free morpheme: Exist on its own
e.g. free roots

Compound: ≥ 2 roots & can be written as 2 orthographic words

Morphological Trees



Lexeme: Same root e.g. walk, walking

Word Formation

Conversion: Δ word class, w/o Δ morphemes
e.g. Google it

Backformation: Words that aren't made of morphemes but appear so
e.g. hamburger

Clipping: Cutting off words
e.g. public house → pub

↳ **Blending**: Combining clips
e.g. brunch

Acronym: Read as a word

Initialism: Read as its initials

Syntax

* indicates syntax invalid sentence

Constituent Test

Topicalization: Move to front of sentence

cleft: Add "it was ___ that"
Pseudocleft: Put the phrase after "what the ___ did was ..."
"___ is what ..."

Substitution: Replace w/ another constituent

Deletion: Remove the phrase

Passing ≥ 1 is a strong indicator

Noun Phrases

(Det) (Adj)* N (PP)
"an" "the" ≥ 0 adjectives root noun prepositional phrase
"very"

() = optional

* = repeatable

Additions

Low vowels are in center of chart

↳ Tense: [a][e][i][o][u]

Contrastive Distribution: 2 phones occur in same phonological env (Minimal pairs)

Complementary Distribution: 2 phones never in same phonological env (Allophones)

Free Variation: Same phonological env (simply diff pronunciation)

Complex Onset: Onset with 2 or more consonants

Finals Notes

Syntax II

Verb Phrases

V (NP) (NP) (PP)

Prepositional Phrase

P NP

Sentence

NP VP

Types of Verbs

"You can try"

auxiliary verb main verb

Intransitive verb: 1 argument (subj)

Transitive Verb: 2 arguments

Ditransitive verb: 3 arguments

↳ Type of objects

"She gave him a cat"

Indirect (can occur in a PP) Direct

Adjunct: Unnecessary constituents

↳ Argument vs Adjunct:

Remove & see if grammatical & broad meaning const

Word Order

42% SVO 3% VOS

45% SOV 1% OVS

10% VSO < 1% OSV

Inflectional: Word tells you grammatical info (tense, det, etc)

Analytic Language: Rely on strict word order to indicate grammatical rls

Synthetic Language: Flexible word order (Most languages)

Semantics

Reference: Real world referent

Sense: How it is referred

Truth-value gap: Neither true/false
~ vacuously true

Non-Referring Expr: No real-world referent

Constant Ref: Same ref regardless of when/where

↳ e.g. speed of light

Variable Ref: Depends on context

↳ e.g. the lights

Homophones: Pronounce same way

Homographs: Spelt same way

Homonyms: Pronounced & spelt same way

Polysemes: ~ homonym but words have related meanings

Hyponym (under): Subset of another word

Hypernym (over): Superset of another word

Synonyms: Similar meaning

Antonyms: Opp meanings

- gradable "short-tall" (continuum)
- complementary (category) "alive-dead"
- relational (w/each other) "parent-child"

Semantic Shifts

Narrowing

Broadening — e.g. genericization (Google → internet search)

Upgrading — amelioration

Downgrading — pejoration

Entailment — $A \Rightarrow B$

Contradicts — $A = B'$

Paraphrase — $A \Leftrightarrow B$

Presuppose — $(A \Rightarrow B) \wedge (A' \Rightarrow B)$
(implies regardless)

Pragmatics

Locution speaker's words

Illocution Intention

Perlocution Effect

Performatives

Explicit — performative verb

Formulaic — Phrase $\Rightarrow$ performative (conventional phrases)
(e.g. Thanks)

Implicit — Request a performative

Felicitous — Word vs intended action ✓

Infelicitous — ✗ authority, ✗ context, ✗ sincerity

Classification of Speech Acts

Representatives — Represent a belief

Directives — Instructions

Commissives — Commit to action

Expressives — Emotion

Declarations — Declare & state

Gricean Maxims

Quality — Truth

Quantity — Amt of info

Relation — Relevance

Manner — Ambiguity & fillers

Flouting a maxim generates extra meaning known as conversational implicature

Politeness Theory

Positive face/politeness: Desire to be liked

Negative face/politeness: Desire for autonomy

Sociolinguistics

Dialect: Δ phonology, lexicon, syntax

Accent: Δ phonetics & phonology

Mutual intelligibility: Both understand each other

Factors on language use

Linguistic: linguistic env

Social: Social background (age, gender, etc)

Stylistic: Δ in situation, although same speaker

Standard language: $\uparrow$ status

Non-standard language: $\uparrow$ solidarity/friendliness

Types of code switching

Inter-sentential — Δ lang. / sentences / clauses

Intra-sentential — Δ lang within clause/sentence

Tag-switching — Δ in some discourse

Pidgins: No native speakers

Creoles: Contact lang w/ native speakers

↳ 1 superstrate, multiple primary substrates

Psycholinguistics

Perceptual narrowing:

Children specialize in contrastive phonemes in their language within 1st year
↳ ✗ react to speech sounds not in language

Babbling: 6 months

First word: 1 year

fis phenomenon:

- Can hear wrongly pronounced phonemes
- Comprehension before production

Critical period: Limited window to acquire language & reach native level

Universal Grammar: Innate, universal grammatical principles (poverty of the stimulus)

Brain:

↳ Language mostly in left hemisphere but right still important

↳ Broca's area: Production

↳ Wernicke's area: Comprehension

Aphasia: Language impairment, depending on certain areas in brain

Linguistic Determinism: Language limits thought

Linguistic Relativism: Language influences thought

Linguistic Universalism: Language ✗ thought

Thinking for speaking: Language shapes the way we prepare thoughts to be expressed

Connectionist: Neural network idea